

AI ENGINEER CAREER ROADMAP

Student -> AI Engineer

Personalized 2-Year Strategy | Built with Chain-of-Thought Reasoning | 2024 - 2026

CURRENT POSITION

- Status: Student
- Core Skill: Python programming
- Strengths: Analytical thinking, coding base
- Stage: Foundation Builder (ready to grow)

TARGET GOAL

- Role: AI / ML Engineer (industry-ready)
- Timeline: 24 months from today
- Outcome: Full-time AI engineering role
- Industries: Tech, Fintech, HealthTech, Research

SKILL GAP ANALYSIS

You Have

- Python (solid base)
- Logical reasoning
- Learner's mindset

Must Learn

- ML/DL Frameworks
- Linear Algebra & Stats
- LLMs & Prompt Eng.

Tools Needed

- PyTorch / TensorFlow
- HuggingFace, LangChain
- Docker, Git, AWS/GCP

Soft Skills

- Systems design thinking
- Technical writing
- Open-source mindset

RECOMMENDED LEARNING PLAN

Phase 1

Months 1-6

- Math for ML (fast.ai)
- NumPy, Pandas, Matplotlib
- Scikit-learn fundamentals
- Kaggle beginner competitions

Phase 2

Months 7-12

- PyTorch deep dive
- CNNs, RNNs, Transformers
- HuggingFace pipelines
- End-to-end ML project

Phase 3

Months 13-18

- LLMs & RAG pipelines
- LangChain, LlamaIndex
- MLOps: MLflow, Docker
- Deploy model to cloud

Phase 4

Months 19-24

- AI system design
- Open-source contributions
- Build & ship AI product
- Interview prep + apply

SUGGESTED PROJECTS

- Beginner** Movie recommender (Scikit-learn)
- Mid-level** Sentiment analyser (fine-tune BERT)
- Advanced** RAG chatbot on custom documents
- Capstone** AI SaaS tool — deploy & monetise

NETWORKING STRATEGY

- LinkedIn:** Post weekly AI learnings & project demos
- GitHub:** Commit daily — build a strong green graph
- Kaggle:** Compete + engage in discussions & forums
- Twitter/X:** Follow AI researchers, share insight threads
- Communities** Discord (AI/ML), Reddit r/MachineLearning

MONTHLY MILESTONES

- M 1-3** Complete ML math + Scikit-learn; push 1st GitHub project
- M 4-6** Finish PyTorch; win/submit 1 Kaggle challenge
- M 7-9** Fine-tune Transformer; grow GitHub to 5+ repos
- M 10-12** Deploy full ML app; reach 500+ LinkedIn followers
- M 13-18** Master LLMs & RAG; launch public AI side-project
- M 19-24** Open-source contribution; apply for AI roles; land offer

IMMEDIATE NEXT ACTIONS — Do This Week

- Enroll in fast.ai 'Practical Deep Learning' (free)
- Create GitHub profile — push a Python project today
- Set up LinkedIn — headline: 'Aspiring AI Engineer | Python'
- Join Kaggle — attempt any beginner competition now
- Install PyTorch & complete the official 60-min blitz tutorial
- Subscribe to 'The Batch' newsletter by DeepLearning.AI